Business Problem Statement

A retail company aims to gain deeper insights into customer shopping behavior to enhance sales performance, improve customer satisfaction, and strengthen long-term loyalty. The organization has observed noticeable shifts in purchasing patterns across different demographics, product categories, and sales channels, including both online and offline platforms.

The management is particularly interested in understanding the key factors influencing customer decisions, such as pricing strategies, discounts, product reviews, seasonal trends, and preferred payment methods. Identifying these drivers is essential for improving customer engagement and optimizing overall business strategies.

The primary objective of this project is to analyze customer behavior data to answer the following business question:

"How can consumer shopping data be leveraged to identify trends, enhance customer engagement, and optimize marketing and product strategies?"

Deliverables

1. Data Preparation & Modeling (Python)

- Cleaned and transformed raw data for analysis
- Handled missing values and inconsistencies
- Prepared structured datasets for further processing

2. Data Analysis (SQL)

- Organized data into a structured format
- Performed queries to analyze customer segments and purchasing behavior
- Identified trends related to customer loyalty and key purchase drivers

3. Visualization & Insights (Power BI)

- Developed an interactive dashboard
- Visualized key metrics, trends, and behavioral patterns
- Enabled stakeholders to interpret data effectively

4. Report & Presentation

- Documented findings and insights in a structured report

- Created a presentation to communicate results and recommendations clearly

5. GitHub Repository

- Maintained a well-structured repository
- Included Python scripts, SQL queries, and dashboard outputs
- Ensured clarity and organization for easy understanding and reuse